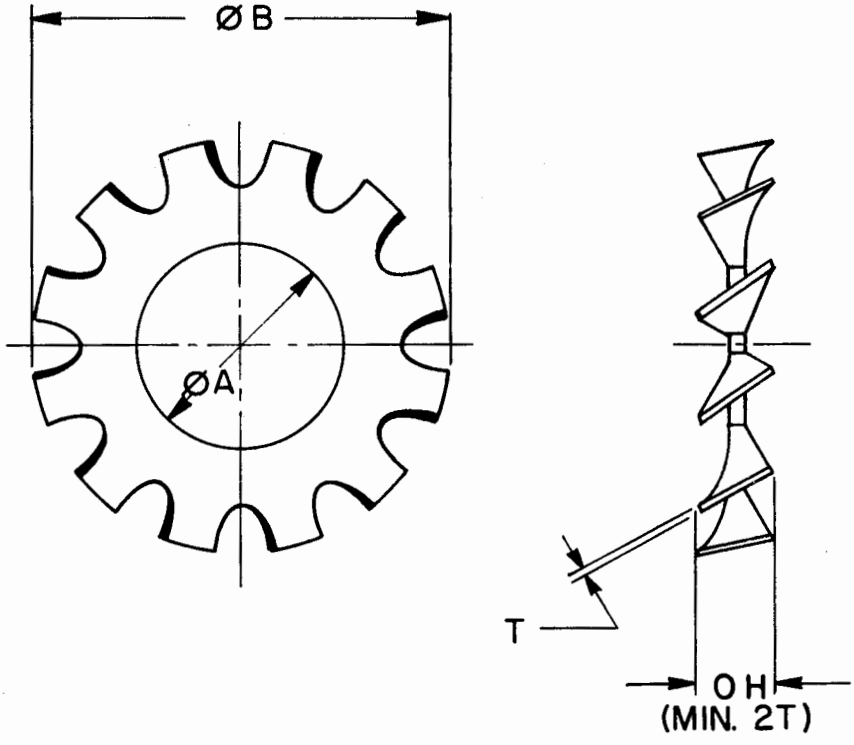
MILITARY SPECIFICATION SHEET

WASHER, LOCK, FLAT, EXTERNAL TOOTH (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the washers described herein shall consist of this document and the latest issue of DOD-W-70336.




NOTE:

1. All dimensions are in millimeters, see Table I for dash numbers and dimensions.

FIGURE 1. WASHER, LOCK, FLAT, EXTERNAL TOOTH (METRIC).



DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:23Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

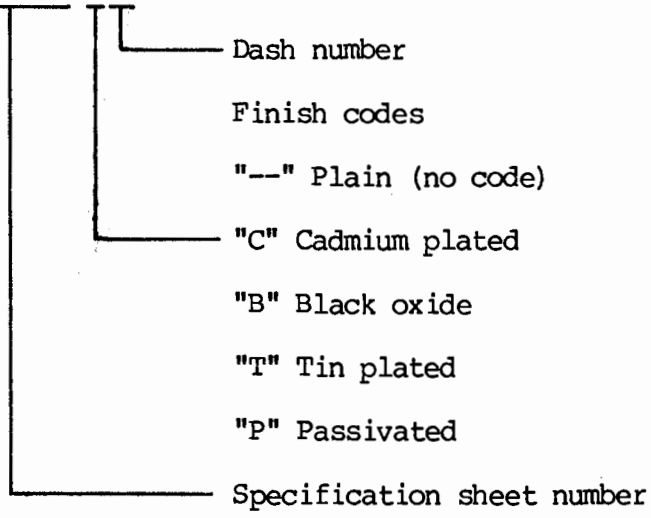
1. Material. Carbon steel, 1050 thru 1064 (UNS G10500 thru UNS G10640), in accordance with ASTM A510M phosphor bronze, (UNS C51000), temper H04 in accordance with ASTM B139M, and austenitic corrosion resistant steel (CRES) 300 Series, non-magnetic in accordance with QQ-S-763.

2. Surface hardness. Carbon Steel washers shall have a hardness of 40-50 HRC. Phosphor bronze washers shall have a hardness of 85-95 HRB. Austenitic (CRES) washer shall have a hardness of 35-43 HRC.

3. Protective coating and surface teratment. Carbon steel washers shall be cadmium plated (Code P) in accordance with QQ-P-416, Type II, Class 2. Phosphor bronze washers shall be tin plated (Code T) in accordance with MIL-T-10727, Type I. Carbon steel and phosphor bronze washers may be plain (no code), with a light film of oil or rust inhibitor. CRES washers shall be passivated (Code P) in accordance with ASTM A380, or black oxide coated (Code B) in accordance with MIL-C-13924.

4. Part number. The part number consists of the letter "D" the basic number of this specification sheet, the finish code letter, and a dash number taken from Table I.

EXAMPLE: D70336/4-C01




D70336/4-C01 Indicates - Washer, Lock, Flat, External Tooth; Carbon Steel; Cadmium Plated; 1.7 mm ID; 3.6 mm OD; 0.3 mm thick.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:23Z
Check the source to verify that this is the current version before use.

TABLE I. DASH NUMBERS AND CHARACTERISTICS.

| NOM SIZE | DASH NO. | | | Ø A ID | | Ø B OD | | THICKNESS T | | FREE HEIGHT OH (REFERENCE) (2T, MIN) | NO. OF TEETH (MIN) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | CARBON STEEL | PHOSPHOR BRONZE | CRES | | TOL | | TOL | | TOL | | |
| 1.6 | 01 | 21 | 41 | 1.7 | | 3.6 | | 0.3 | | 0.6 | 6 |
| 1.8 | 02 | 22 | 42 | 1.8 | | 3.8 | 0 | 0.3 | ±.02 | 0.6 | 6 |
| 2 | 03 | 23 | 43 | 2.2 | +0.08 | 4.5 | -0.08 | 0.3 | | 0.6 | 6 |
| 2.5 | 04 | 24 | 44 | 2.7 | 0 | 5.5 | | 0.4 | ±.03 | 0.8 | 6 |
| 3 | 05 | 25 | 45 | 3.2 | | 6 | 0 | 0.4 | | 0.8 | 6 |
| 3.5 | 06 | 26 | 46 | 3.7 | | 7 | -0.12 | 0.5 | | 1 | 6 |
| 4 | 07 | 27 | 47 | 4.3 | | 8 | | 0.5 | ±.04 | 1 | 8 |
| 5 | 08 | 28 | 48 | 5.3 | +0.12 | 10 | 0 | 0.6 | | 1.2 | 8 |
| 6 | 09 | 29 | 49 | 6.4 | 0 | 11 | -0.15 | 0.7 | ±.05 | 1.4 | 8 |
| 8 | 10 | 30 | 50 | 8.4 | +0.15 | 15 | 0 | 0.8 | | 1.6 | 8 |
| 10 | 11 | 31 | 51 | 10.5 | 0 | 18 | -0.18 | 0.9 | | 1.8 | 9 |
| 12 | 12 | 32 | 52 | 12.5 | | 20.5 | | 1 | ±.06 | 2 | 10 |
| 14 | 13 | 33 | 53 | 14.5 | +0.18 | 24 | | 1 | | 2 | 10 |
| 16 | 14 | 34 | 54 | 16.5 | 0 | 26 | 0 | 1.2 | | 2.4 | 12 |
| 18 | 15 | 35 | 55 | 19 | | 30 | -0.21 | 1.4 | ±.08 | 2.8 | 12 |
| 20 | 16 | 36 | 56 | 21 | +0.21 | 33 | | 1.4 | | 2.8 | 12 |
| | | | | | 0 | | | | | | |





Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:23Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AR
Navy - OS
Air Force - 99

Review Activities:
Army - AV, EA, MI
Navy - SH
Air Force - 82
DLA - IS
NSA - NS

User Activities:
Army - AT, CR, ER, ME
Navy - MC.

Preparing Activity:
Army - AR

Agent:
DLA - IS

(Project 5310-1551)



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:23Z
Check the source to verify that this is the current version before use.